

LLM Based Historical Accounting Evaluation: Abraham Lincoln vs. Historical Accounts

ML Evaluation Engineer Technical Assessment

Sanket Deshpande
pvdsan@gmail.com
Memory Machines

December 6, 2025

Contents

1	Data Acquisition and Normalization	3
1.1	Data Sources	3
1.2	Library of Congress Acquisition	3
1.3	Normalization Schema	3
1.4	Transformation Layer	4
1.4.1	Encoding Fixes	4
1.4.2	Metadata Extraction	4
1.4.3	Name Normalization	4
1.5	Summary	4
2	Event Extraction and Evaluation Pipeline	5
2.1	Overall Architecture	5
2.2	Hybrid Search Components	5
2.3	Performance Comparison	7
2.4	Rationale for Selecting Strategy A: Temporal Confusion in RAG	7
2.4.1	Case Study: Election Night 1860 vs. 1864	7
2.4.2	Root Cause and Conclusion	7
3	LLM Judge	8
3.1	Comparison Task	8
3.2	Scoring Rubric	8
3.3	Prompting Strategies	8
3.4	Events Evaluated	8
4	Statistical Validation	9
4.1	Experiment 1: Prompt Robustness (Ablation Study)	9
4.1.1	Integrated Observations: Prompt Stability and Author Behavior	9
4.2	Experiment 2: Self-Consistency (Reliability Across Runs)	9
4.2.1	Unreliable Comparison Pairs	10
4.3	Experiment 3: Inter-Rater Agreement (Human Alignment)	10
4.3.1	The CoT Kappa Problem	11

1 Data Acquisition and Normalization

This section summarizes the data engineering pipeline used to ingest and normalize primary and secondary historical sources for downstream LLM extraction and evaluation.

1.1 Data Sources

We collect two complementary corpora:

- **Primary Sources (Library of Congress):** Five first-person documents authored by Abraham Lincoln (letters, speeches). These serve as factual ground truth.
- **Secondary Sources (Project Gutenberg):** Five full-length biographies representing historian interpretations. These provide the comparison set for consistency scoring.

1.2 Library of Congress Acquisition

The LoC exposes documents through heterogeneous endpoints. We implement an automatic URL classifier to select the appropriate extraction method.

Table 1: LoC URL types and their extraction paths.

URL Pattern	Example	Method
/item/{id}	loc.gov/item/mal4361800	JSON API
/resource/{id}	loc.gov/resource/mal.4361800	JSON API
/exhibits/...	loc.gov/exhibits/...	HTML Scraping

Extraction logic:

1. Use the undocumented JSON API (?fo=json) for all item/resource URLs.
2. For exhibit pages, scrape the DOM content block with BeautifulSoup.

A conservative 3.5s delay ensures compliance with LoC rate limits.

Table 2: Primary documents retrieved from the Library of Congress.

Document	Type	Format
Letter to Truman Smith (1860)	Letter	PDF + Transcription
Fort Sumter Decision	Report	OCR Text
Gettysburg Address (Nicolay Copy)	Speech	HTML
Second Inaugural Address	Speech	Transcription
March 1865 Correspondence	Letter	PDF

1.3 Normalization Schema

All documents—regardless of source—are converted into a single canonical JSON representation:

```
1 {  
2   "id": "uuid_v4",  
3   "title": "Gettysburg Address",  
4   "reference": "LoC-mal.4361300",  
5   "document_type": "Speech",  
6   "date": "1863-11-19",  
7   "content": "Four score and seven years ago..."  
8 }
```

Listing 1: Unified Normalized Schema

This enables identical downstream processing for both primary and secondary sources.

1.4 Transformation Layer

The transformation layer performs minimal but essential corrections to ensure clean, machine-consumable text.

1.4.1 Encoding Fixes

OCR and legacy encodings introduce consistent corruption patterns. A substitution dictionary resolves them:

Table 3: Common encoding corrections applied to LoC transcriptions.

Corrupted	Correct
â€œ	"
â€™	'
â€"	—
Ã©	é

1.4.2 Metadata Extraction

LoC documents include inconsistent headers. A lightweight regex parser extracts title, date, participants, and location, with fallback to API metadata.

1.4.3 Name Normalization

Names across biographies vary in formatting (suffixes, titles, CSV-style authorship). We canonicalize them via deterministic rules:

Table 4: Name normalization examples.

Raw	Normalized
Lincoln, Abraham	Abraham Lincoln
Morse, John T., Jr.	John T. Morse Jr.
Charnwood, Godfrey Rathbone Benson, Baron	Baron Godfrey Rathbone Benson Charnwood
Hay, John, Nicolay, John G.	John Hay; John G. Nicolay

—

1.5 Summary

This pipeline converts heterogeneous historical documents into a clean, unified, metadata-rich format suitable for reliable LLM-based extraction. The normalization layer ensures that downstream differences between primary and secondary narratives reflect genuine historiographical variance rather than noise from inconsistent document formatting.

2 Event Extraction and Evaluation Pipeline

This section outlines the full pipeline for extracting historically grounded event–claim structures from primary sources (Library of Congress letters and speeches) and long-form secondary biographies (Project Gutenberg). The system supports both short and long documents, and incorporates two alternative strategies for scalable extraction from book-length texts.

2.1 Overall Architecture

The extraction framework operates in two modes depending on document length and structure:

Short-Document Mode (Library of Congress). Primary-source materials are typically concise and fit within a single LLM context window. These documents are processed in a single forward pass using a dedicated `prompt_direct.md` template, which extracts all five target events—claims, timestamps, and tonal descriptors. This mode provides clean, high-precision reference data and serves as an upper bound on extraction quality.

Long-Document Mode (Gutenberg Biographies). Full-length biographies exceed LLM context limits and require more sophisticated processing. We evaluate two competing strategies:

- **Strategy A: Exhaustive Batch Processing** A deterministic, full-coverage approach that segments each book into overlapping 30k-token windows. Every batch is analyzed independently for mentions of the five target events, and results are later consolidated into a unified event record.
- **Strategy B: Hybrid Retrieval-Augmented Generation (RAG)** A targeted extraction method combining semantic retrieval (FAISS) and lexical retrieval (BM25). Instead of processing the entire book, only the top-ranked chunks are passed to the LLM, dramatically reducing API usage.

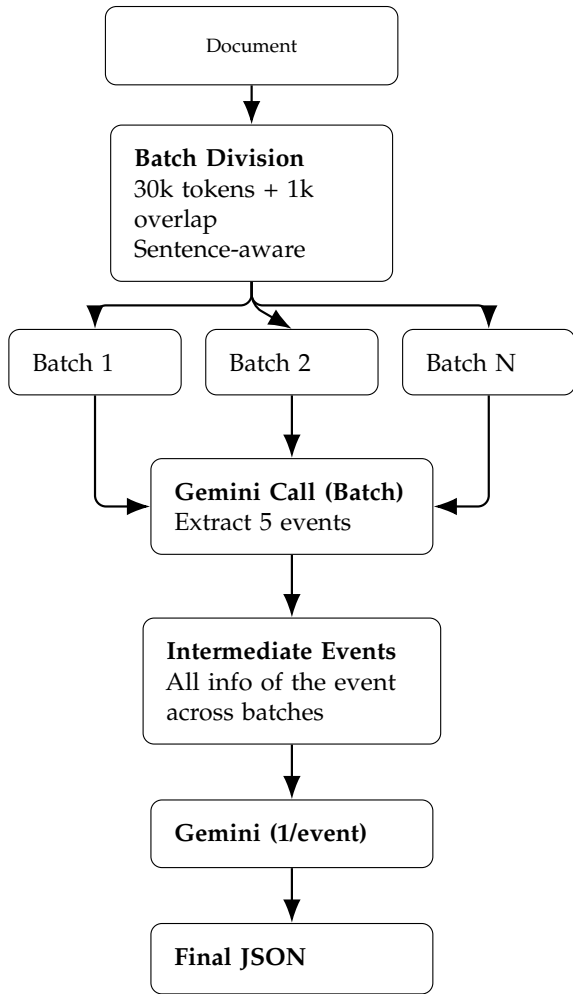
Figure 1 visualizes both long-document strategies. Strategy A prioritizes recall and narrative continuity, whereas Strategy B prioritizes efficiency through targeted retrieval. Subsequent sections analyze their empirical behavior and motivate the final system choice.

2.2 Hybrid Search Components

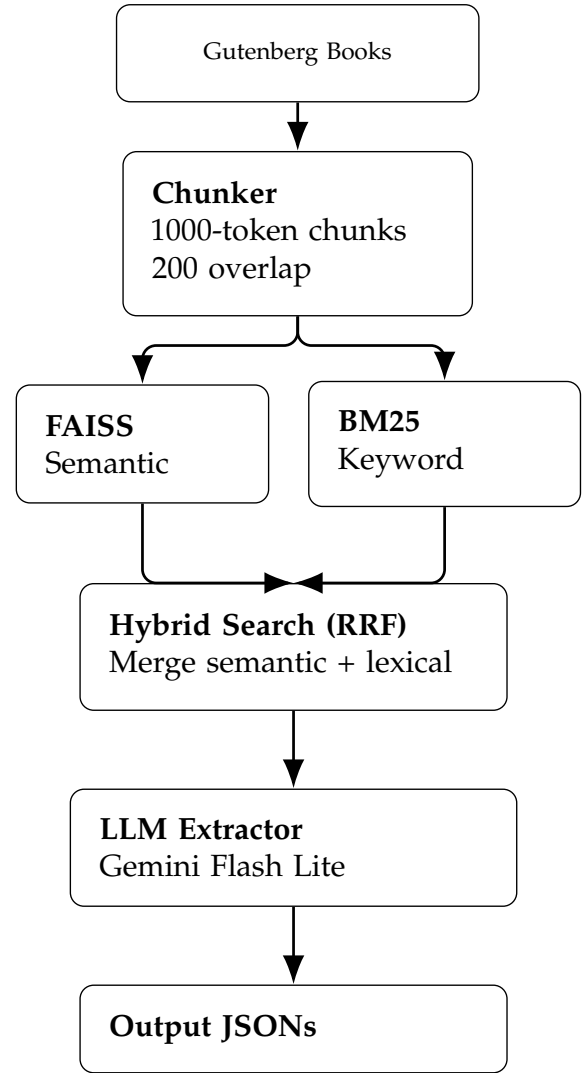
Table 5: Configuration of the Sentence-Embedding Index

Property	Value
Embedding Model	all-MiniLM-L6-v2
Dimension	384
Index Type	Local FAISS (Inner Product)
Throughput	~1000 embeddings/sec

Indices. FAISS provides dense semantic similarity, while BM25 supplies high-precision keyword matching. RRF blends the rankings to mitigate failure modes of either individual index.



(a) **Strategy A — Batch Extraction and Consolidation.** The document is divided into sentence-aware 30k-token batches with overlap to preserve continuity. Each batch is processed independently using a dedicated extraction prompt, producing normalized JSON fragments. Intermediate event-level files are merged across all batches before a final Gemini pass produces fully structured outputs (claim sets, temporal metadata, and tonal annotations). This strategy prioritizes comprehensive coverage of the entire text corpus, ensuring no events are missed due to retrieval bias.



(b) **Strategy B — Hybrid RAG Retrieval + Targeted Extraction.** Long documents are chunked into overlapping semantic units and indexed using both a FAISS vector index (embedding-based similarity) and a BM25 lexical index. A hybrid retrieval stage combines both ranking signals via Reciprocal Rank Fusion (RRF), ensuring that semantically relevant and keyword-matching passages are jointly prioritized. Only the top-ranked chunks are passed to the extraction model, allowing Gemini to operate on a focused subset of highly relevant text. This approach reduces computation and latency while maintaining high precision in event-level extraction.

Figure 1: Side-by-side comparison of the two extraction strategies. Strategy A provides complete-document coverage via batching, whereas Strategy B retrieves only the most relevant chunks using hybrid semantic-lexical search.

2.3 Performance Comparison

Table 6: API Usage Breakdown for Both Strategies

Method	Source	Stage	Formula	Calls
Strategy A	LoC	Direct Extraction	5×1	5
	Gutenberg	Batch Stage	5×4 batches	~ 20
	Gutenberg	Consolidation	5×5 events	25
		Total		~ 50
Strategy B	LoC	Direct Extraction	5×1	5
	Gutenberg	Retrieval	Local FAISS + BM25	0
	Gutenberg	Event Extraction	5×5 events	25
		Total		~ 30

2.4 Rationale for Selecting Strategy A: Temporal Confusion in RAG

Despite lower API usage, Strategy B suffered from a consistent and severe failure mode: **Temporal Confusion**. Semantic retrieval mistakenly merged narratives from distinct historical periods when keyword distributions aligned.

2.4.1 Case Study: Election Night 1860 vs. 1864

Embedding similarity caused Strategy B to retrieve 1864 election passages for queries related to 1860, as shown in Table 7. This misalignment propagated into extracted claims, corrupting the factual correctness of downstream consistency scoring.

Table 7: Example of RAG-Induced Temporal Confusion

Feature	RAG Output (Incorrect)	True 1860 Account
Location	War Department, Washington	Springfield, Illinois
Key Figures	Edwin Stanton	Stanton not yet an ally
Lincoln’s Role	Sitting President	Presidential Candidate
Source Event	1864 Election	1860 Election

2.4.2 Root Cause and Conclusion

Chunk-based retrieval removes the narrative chronology required to distinguish recurring historical events (e.g., “Election Night”). Because embeddings encode semantic similarity but not event temporality, RAG systematically conflated time-separated passages.

By contrast, Strategy A preserves narration order, allowing the model to track evolving historical context (e.g., Stanton only appears post-1862). Although more costly, Strategy A guarantees factual reliability—a critical requirement for downstream consistency scoring.

We therefore adopt Strategy A as the production-standard extraction pipeline.

3 LLM Judge

3.1 Comparison Task

For each event, the LLM Judge answers:

“How consistent is Author X’s account with Lincoln’s own words?”

The evaluation pair consists of (1) factual claims extracted from Lincoln’s speeches and letters, and (2) interpretive statements from the corresponding biographer.

The output is a numerical score (0–100) mapped to the rubric in Table 8.

3.2 Scoring Rubric

Range	Label	Meaning
90–100	High Fidelity	Matches Lincoln closely; additions are factual context.
70–89	Contextual Variance	Facts align, but narrative framing diverges meaningfully.
50–69	Interpretive Drift	Minor factual or tonal inconsistencies.
0–49	Contradiction	Clear conflict or misrepresentation of Lincoln.

Table 8: Consistency rubric used across all prompting strategies.

3.3 Prompting Strategies

We evaluate three commonly used prompting paradigms:

- **Zero-Shot:** Instructions + rubric only. Lightweight but prone to higher variance due to missing calibration anchors.
- **Chain-of-Thought (CoT):** Enforces a reasoning sequence

Fact Matching → Context → Tone → Score.

Produces structured explanations but may collapse into majority-class predictions.

- **Few-Shot:** Includes three scored examples (non-Lincoln cases). Helps stabilize output formatting and calibration, but introduces example-induced bias.

3.4 Events Evaluated

Four events were included in the robustness study, selected to span a wide range of evidence density (1–22 primary claims). This allows us to observe how prompt sensitivity changes when the factual grounding varies.

Event	# Claims	# Authors	Included?
Election Night 1860	1	5	✓
Fort Sumter Decision	4	5	✓
Gettysburg Address	10	5	✓
Second Inaugural Address	22	5	✓
Ford’s Theatre Assassination	0	5	× ^a

^a Excluded because Lincoln provided no primary claims about his own assassination, making consistency evaluation impossible.

Table 9: Events included in the prompt robustness study, ordered by evidence density.

4 Statistical Validation

To evaluate the reliability and interpretability of the LLM Judge, we conducted three complementary experiments: (1) prompt robustness, (2) self-consistency, and (3) alignment with human annotators. Together, these quantify stability, variance, and agreement quality.

4.1 Experiment 1: Prompt Robustness (Ablation Study)

This experiment measures how sensitive the LLM Judge is to different prompting strategies.

Strategy	Mean Score	Std Dev	Min	Max	Verdict
Zero-Shot	80.25	13.13	40	95	High Variance
Chain-of-Thought (CoT)	82.50	9.25	70	95	Most Stable
Few-Shot	72.00	21.91	20	95	Unstable

Table 10: Ablation study comparing three prompting strategies. CoT demonstrates the highest stability across repeated evaluations, while Few-Shot exhibits significantly higher variance due to sensitivity to example conditioning.

4.1.1 Integrated Observations: Prompt Stability and Author Behavior

Figure 2 visualizes consistency scores and strategy heatmaps across authors.

Key Findings.

- **Charnwood** shows the highest fidelity (85.83) with minimal drift.
- **Hay and Nicolay** score lowest (65.83), especially under Few-Shot.
- **Ketcham** and **Browne** show stable performance across strategies.
- **Morse** is highly prompt-sensitive, with variance exceeding 17 points.

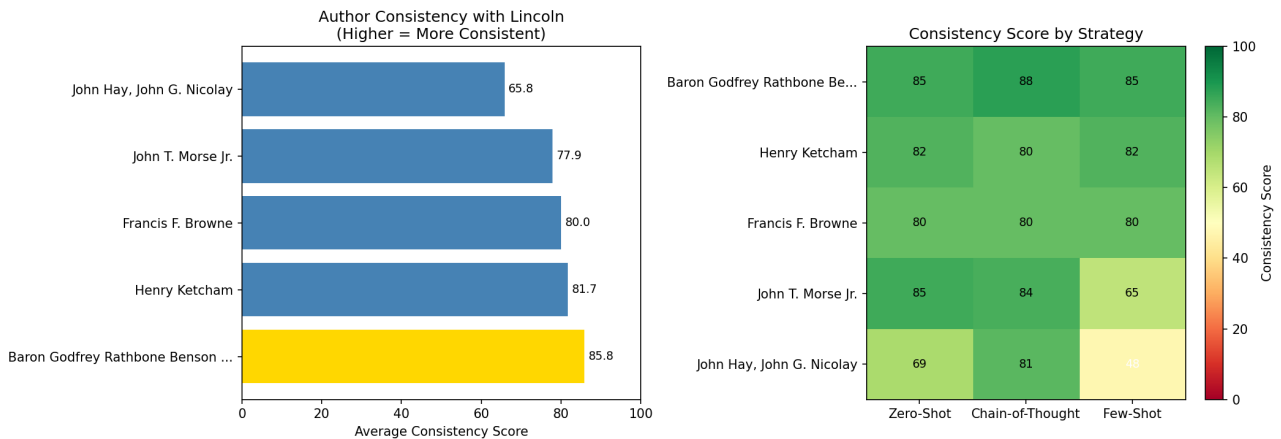


Figure 2: Consistency heatmap for each author across prompting strategies. Left: Mean author–Lincoln consistency. Right: Strategy-wise stability profiles illustrating which authors induce high prompt sensitivity.

4.2 Experiment 2: Self-Consistency (Reliability Across Runs)

Repeated scoring reveals how deterministic each prompting strategy is.

Prompt Strategy	Mean Std Dev	Interpretation
Few-Shot	2.58	Most deterministic
Chain-of-Thought	4.15	Moderate variance
Zero-Shot	4.75	Highest variance

Table 11: **Self-consistency analysis across five independent runs.** Despite instability across pairs, Few-Shot is internally deterministic once examples anchor the evaluation pattern.

4.2.1 Unreliable Comparison Pairs

Figure 3 highlights pairs whose score ranges exceeded 20 points.

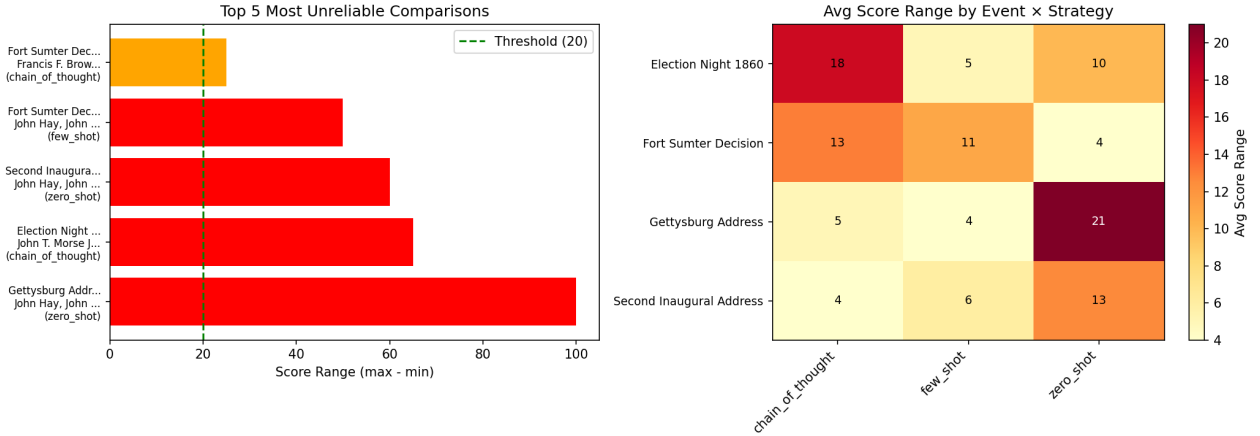


Figure 3: **Unreliable comparison cases.** Left: Top 5 pairs with the largest score ranges. Right: Heatmap showing that high variance clusters around events with few primary-source claims (1–4), while claim-rich events remain stable.

4.3 Experiment 3: Inter-Rater Agreement (Human Alignment)

Procedure

1. A human annotator labeled 10 comparison pairs as **Consistent** or **Not Consistent**.
2. The LLM evaluated the same pairs using all three prompting strategies.
3. Scores were binarized using a 70-point threshold.
4. Cohen’s Kappa was computed to measure agreement beyond chance.

ID	Event	Human	ZS	CoT	FS	ZS Match	CoT Match	FS Match
1	Election Night	Con	75	75	40	✓	✓	×
2	Election Night	Con	75	75	75	✓	✓	✓
3	Fort Sumter	Con	75	75	75	✓	✓	✓
4	Fort Sumter	Con	75	75	75	✓	✓	✓
5	Gettysburg	NotCon	40	75	40	✓	×	✓
6	Gettysburg	Con	95	95	95	✓	✓	✓
7	Second Inaugural	Con	95	95	85	✓	✓	✓
8	Second Inaugural	NotCon	75	80	55	×	×	✓
9	Gettysburg	Con	75	75	75	✓	✓	✓
10	Second Inaugural	Con	95	95	95	✓	✓	✓

Table 12: **Human vs. LLM comparison outcomes.** Checkmarks indicate agreement between human and LLM after binarizing scores at the 70-point threshold.

Strategy	Matches	Agreement	Cohen’s Kappa
Zero-Shot	9/10	90%	0.64
Few-Shot	9/10	90%	0.64
Chain-of-Thought	8/10	80%	0.00

Table 13: **Summary of human–LLM inter-rater agreement.** Zero-Shot and Few-Shot achieve substantial Kappa values, while CoT collapses due to uniform predictions.

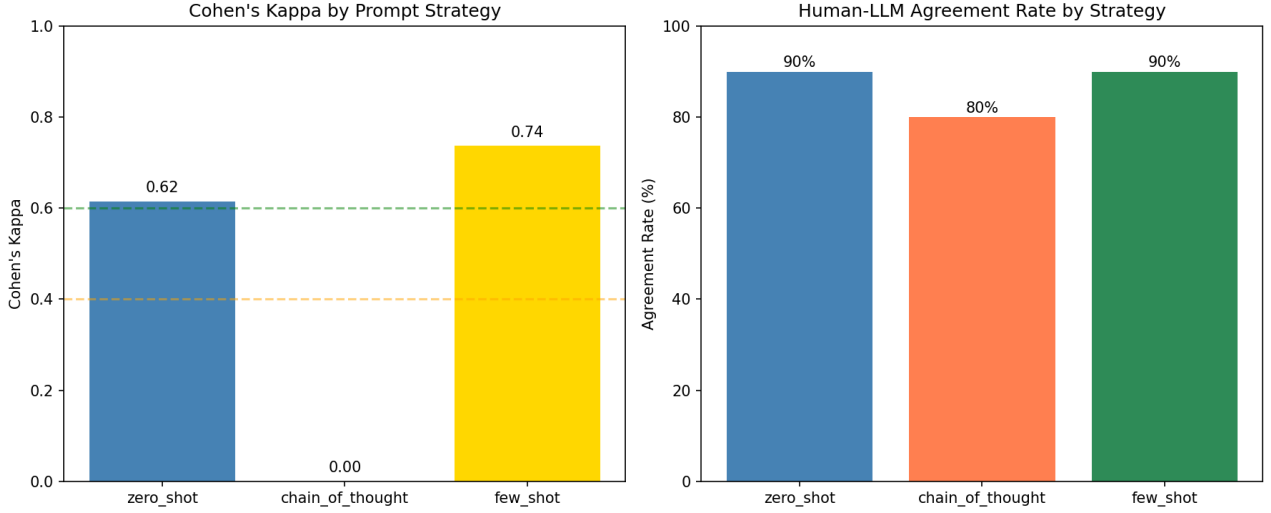


Figure 4: **Inter-rater agreement across prompting strategies.** Left: Cohen’s Kappa scores. Right: Raw human–LLM agreement rates. CoT achieves high raw accuracy but fails to detect “Not Consistent” cases, resulting in $\kappa = 0$. Zero-Shot and Few-Shot exhibit substantial agreement in both raw and chance-corrected metrics.

4.3.1 The CoT Kappa Problem

Although Chain-of-Thought achieved 80% raw agreement, its Kappa score is **0.00** because it predicted all samples as Consistent. This majority-class collapse eliminates discriminative ability, showing that CoT—despite being stable—requires calibration before being used in binary consistency judgments.